

Concrete Example: Constructing the Hamming $[7, 4, 3]$ Cyclic Code

Geometry: The Zero Locus

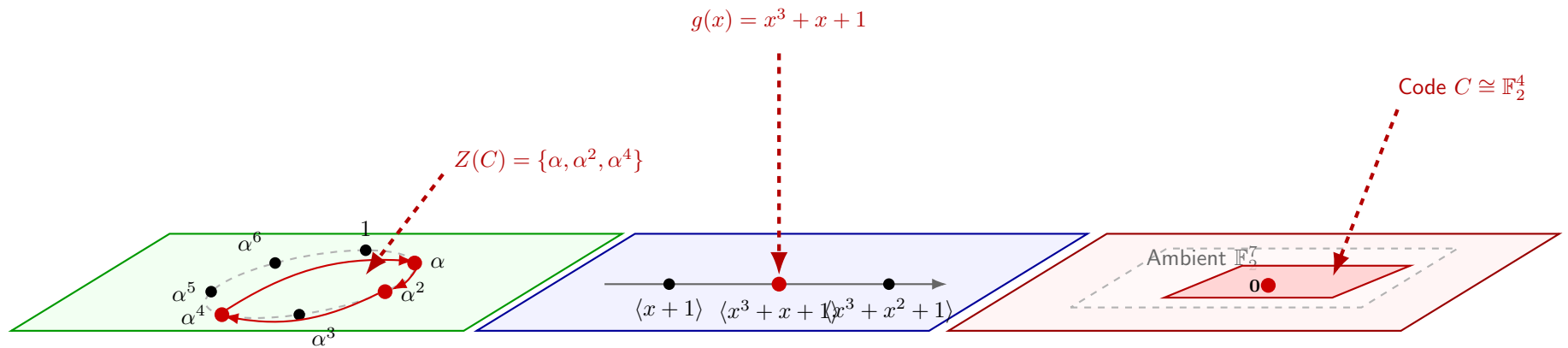
Selecting roots in $\mathbb{F}_2$

Algebra: The Ideal

Generators in $A = \mathbb{F}_2[x]/\langle x^7 - 1 \rangle$

Coding Theory: Subspace

Vector space $\mathbb{F}_2^7$



$$c(\alpha) = 0 \implies c(\alpha^2) = c(\alpha^4) = 0$$

Geometrically, we select a single Frobenius orbit to act as the roots of our codewords.

$$C = \langle g(x) \rangle \subset A$$

Algebraically, the chosen roots are exactly the roots of the irreducible polynomial $g(x)$. The code is the ideal generated by g .

$$Hc^T = \mathbf{0} \quad (\text{Parity Checks})$$

In linear algebra, the roots evaluate to parity check equations. Because $\deg(g) = 3$, it cuts out a $7 - 3 = 4$ dimensional subspace.

To build a cyclic code of length $n = 7$ over $\mathbb{F}_2$:

Select roots $\{\alpha, \alpha^2, \alpha^4\} \iff$ Multiply by $g(x) = x^3 + x + 1 \iff$ Form the 4D Hamming $[7, 4, 3]$ Subspace.